

Numerical Optimization and Large Scale Linear Algebra

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Problem Statement

In this study, we explore different optimization techniques to achieve a **uniform illumination** pattern using a set of $n = 10$ lamps positioned over a 25×25 grid. The illumination across $m = 625$ pixels follows the relationship:

$$\mathbf{l} = \mathbf{A}\mathbf{p}$$

where:

- $(\mathbf{l})$ is the vector of illumination levels across all pixels.
- $(\mathbf{A})$ is an illumination sensitivity matrix capturing how each lamp contributes to each pixel.
- $(\mathbf{p})$ represents the power settings for the lamps.

The goal is to determine **optimal lamp power distributions and placements** to minimize illumination deviations from a desired uniform pattern ($\mathbf{l}_{\text{des}} = \mathbf{1}$).

Question 1: Illumination Changes with Different Power Distributions

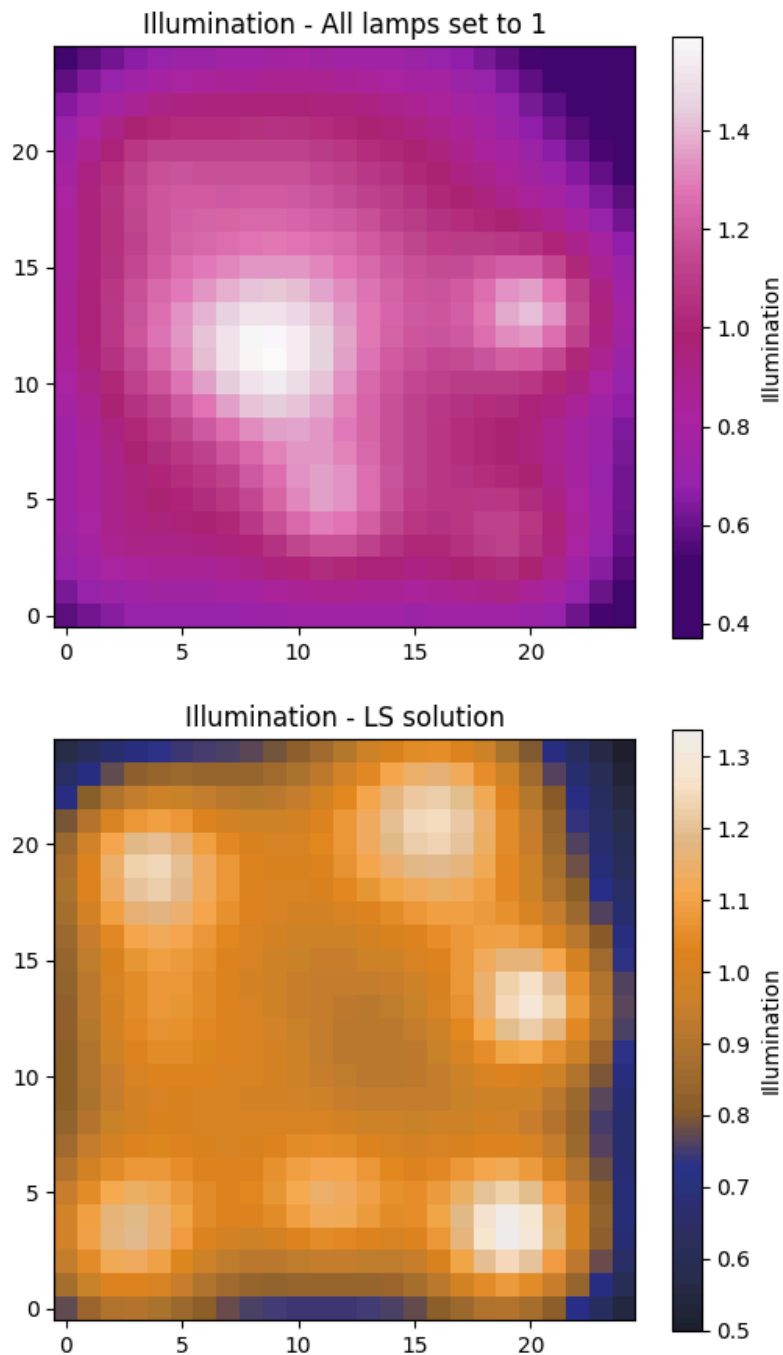
Objective:

- Compare the illumination pattern when **all lamps are set to power 1** against an **optimized least squares (LS) solution**.
- Assess the **Root Mean Squared (RMS) error** to quantify uniformity.

Findings:

- **All Lamps Set to 1:**
 - **RMS Error: 0.2459**
 - Uneven lighting with bright spots near lamps and dim regions further away.
- **Least Squares Solution:**
 - **RMS Error: 0.1458**
 - More **balanced illumination**, but some variations remain.

Conclusion: Least squares optimization significantly **reduces non-uniformity**, though it doesn't eliminate it entirely.



Question 2: Histograms

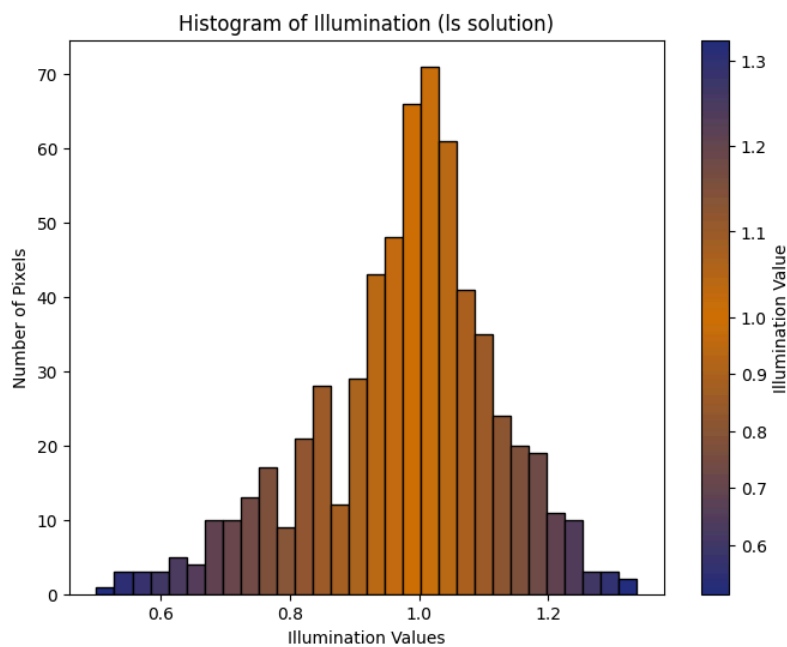
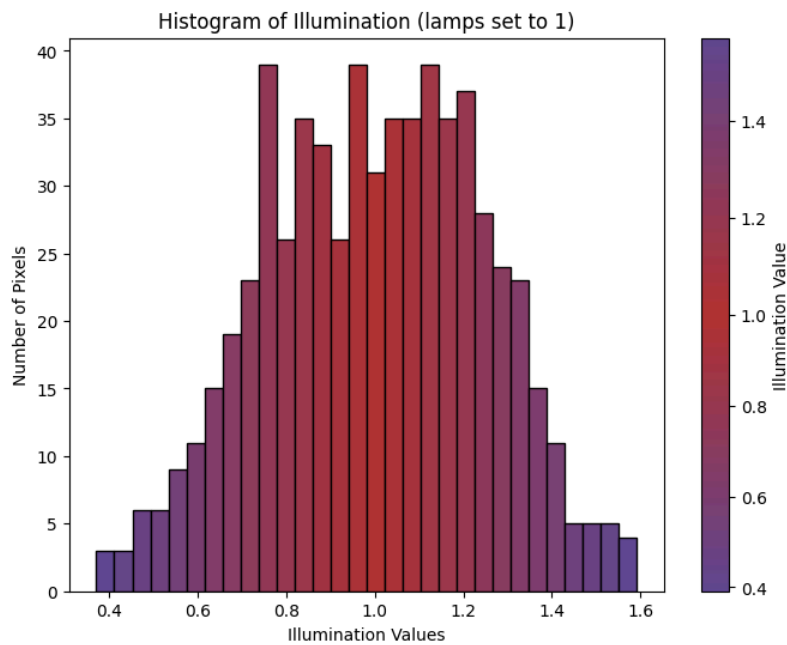
Objective:

- Compare the **illumination distributions** for the two cases from Question 1.

Findings:

- **All Lamps Set to 1:**
 - **Histogram is widely spread**, showing inconsistent illumination across pixels.
- **LS Optimized Powers:**
 - The distribution **narrows**, meaning **illumination values are more concentrated around 1**.

Conclusion: Least squares **reduces variation in illumination**, making lighting more uniform across the area.



Question 3: Add Power Constraints

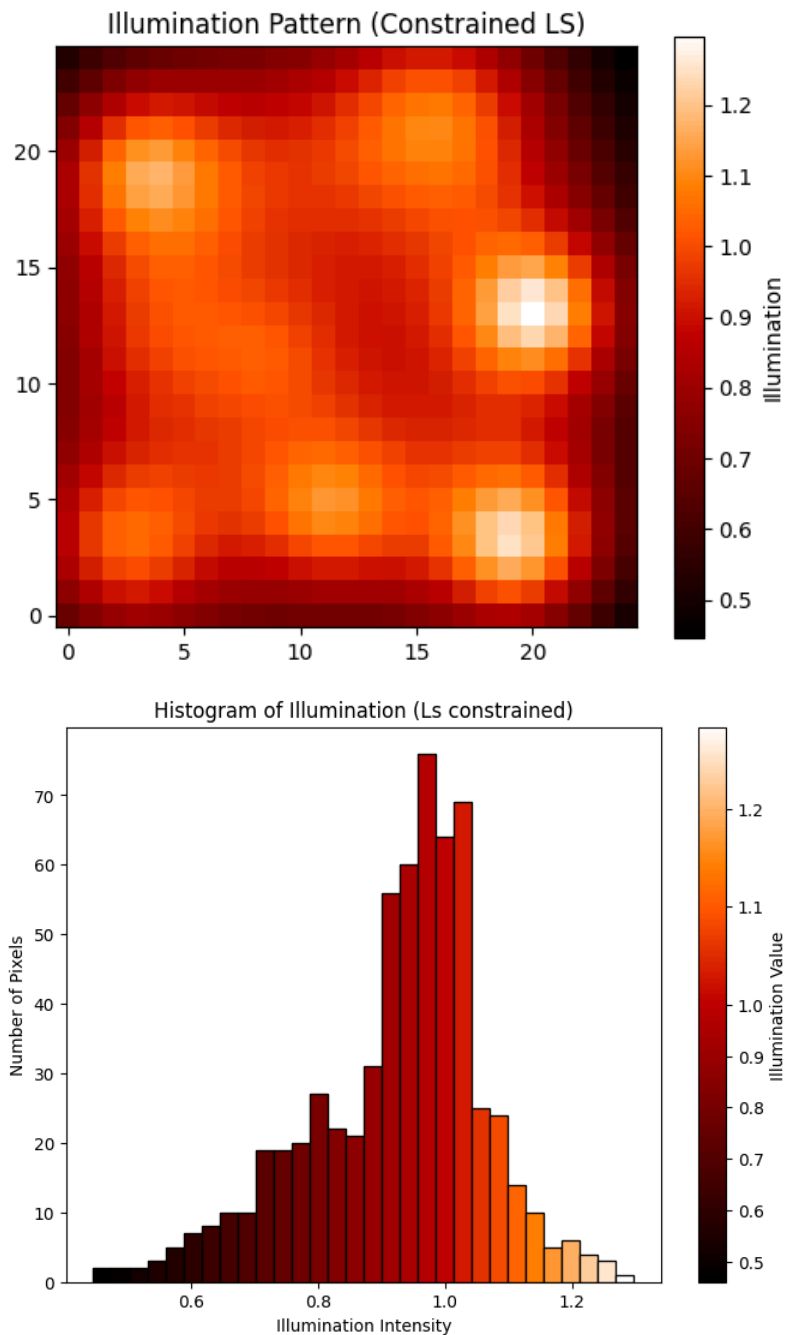
Objective:

- Introduce a **realistic constraint**: Total power across all lamps is **fixed at 10**, and **no lamp can have negative power**.

Findings:

- **Constrained Least Squares Solution**:
 - **RMS Error: 0.1598** (slightly worse than LS due to constraints).
 - The illumination remains **fairly uniform**, but some lamps operate at lower power levels to meet constraints.

Conclusion: The constraints enforce **realistic energy consumption** while maintaining **reasonable illumination uniformity**.



Question 4: Find Even Better Lamp Positions

Objective:

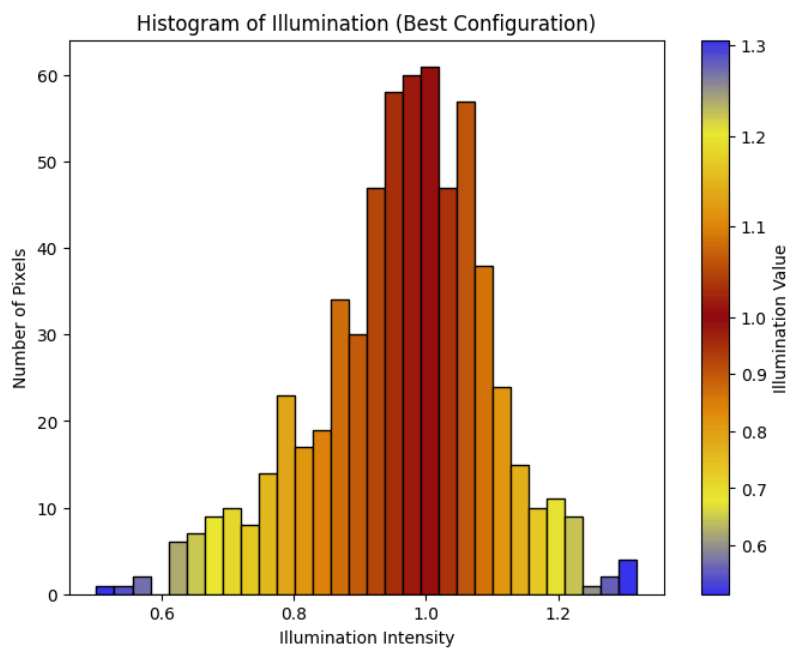
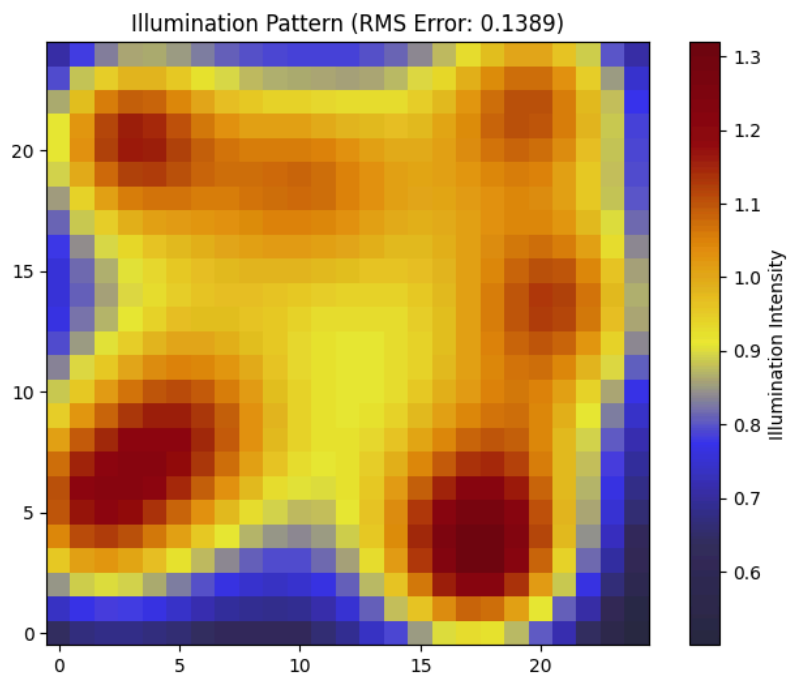
- Optimize **lamp positions** to **further minimize RMS error**.
- Allow **lamp heights between 4 and 6 meters**.
- Ensure the **total lamp power remains 10** and **no lamp has negative power**.

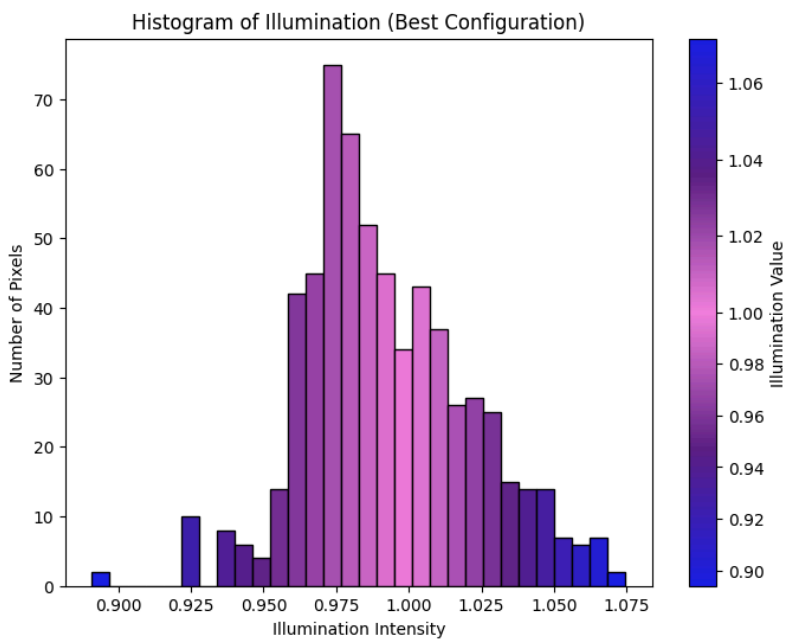
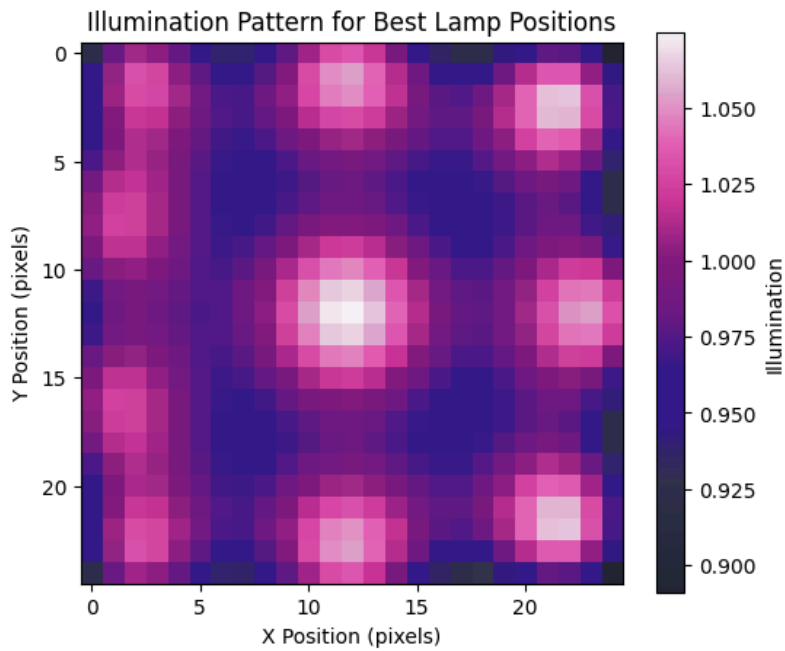
Findings:

- **Random Search Approach:**
 - Found an arrangement with **RMS Error = 0.1389**.
 - Some improvement, but inconsistent results.
- **Differential Evolution (DE) Approach:**
 - **Best RMS Error: 0.0303**
 - Achieved the **most uniform illumination** with a **well-balanced power distribution**.
 - **Fast convergence** with early stopping after **28 iterations**.

Conclusion:

- **Random search** sometimes finds good configurations but is **inefficient**.
- **Differential Evolution** systematically finds optimal placements, outperforming all other methods.





Final Comparison: Which Method is Best?

Method	RMS Error	Illumination Uniformity	Strengths
All Lamps = 1	0.2459	Poor	Simple baseline
Least Squares	0.1458	Moderate	Improves uniformity
Constrained LS	0.1598	Moderate	Enforces power limits
Random Search	0.1389	Inconsistent	Finds better placements sometimes
Differential Evolution	0.0303	Best	Most uniform illumination

Conclusion:

Differential Evolution provides the best illumination uniformity with the lowest RMS error, making it the best approach for lamp optimization.